

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA1407
Design

COURSE NAME: Compiler

COURSE OUTCOME COVERED

CO4: *Examine intermediate code generation techniques to enhance code optimization (BL4)*

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Annexure A

CLASS TEST

Code of Conduct:

*I ___Juhi Surin (192525148)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Juhi Surin

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization (15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CLASS TEST

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

Q1. Write the three address code for the following:
given statement:

if $((a < b) \text{ and } (c < d) \text{ or } (a > d))$ then

$z = x + y * z$

else

$z = z + 1$

38/40

Sol.

condition:

$a < b$

$c < d$

$a > d$

Labels: L1, L2, L3, L4, L5, L6

Three address code for the statement:

① L1:

if $a < b$ goto L2

goto L5

⑤ L5:

$t3 = z + 1$

$z = t3$

② L2: if $c < d$ goto L4

goto L3

⑥ L6:

③ L3: if $(a > d)$ goto L4

goto L5

④ L4:

$t1 = y * z$

$t2 = x + t1$

$z = t2$

goto L6

Q2) Translate $(a < b)$ and $(c < d)$ or $(e < f)$ into three address statements using back patching.

Sol. $[(a < b) \text{ AND } (c < d)] \text{ OR } (e < f)$

1. if $a < b$ goto 3
2. goto 5
3. if $(c < d)$ goto 7
4. goto 5
5. if $(e < f)$ goto 7
6. goto 8
7. True
8. False

Q3) Consider the arithmetic expression:
 $A = (-A + B) * ((-A + B) * c)$

Sol. given statement:

$$A = (-A + B) * ((-A + B) * c)$$

Three Address code:

$$t1 = -A$$

$$t2 = t1 + B$$

$$t3 = -A$$

$$t4 = t3 + B$$

$$t5 = t4 * c$$

$$t6 = t2 * t5$$

$$A = t6$$

Quadruples

<u>Operator</u>	<u>arg 1</u>	<u>arg 2</u>	<u>result</u>
-	A	-	t1
+	t1	B	t2
-	A		t3
+	t3	B	t4
*	t4	C	t5
+	t2	t5	t6
=	t6	-	A

Q4) Construct three address code using the following production for Boolean expression $(a > b) \text{ or } (c < d) \text{ or } (e < f)$.

$B \rightarrow B1 \parallel B2$

$B \rightarrow B1 \&\& B2$

$B \rightarrow !B1$

$B \rightarrow E1 \text{ relop } E2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

Sol. Given expression:

$(a > b) \text{ or } (c < d) \text{ or } (e < f)$

Three Address Code

L1:

if $a > b$ goto LTRUE

goto L2

L2:

if $c < d$ goto LTRUE
goto L3

L3:

if $e < f$ goto LTRUE
goto LFALSE

LTRUE:

// Boolean expression is true
goto LEND

LFALSE:

// Boolean expression is false

LEND:

